MILITARY SPECIFICATION

BATTERIES, PRIMARY, WATER-ACTIVATED

(DUNK TYPE)

This interim amendment is issued for use by the Department of the Army with Military Specification MIL-B-10154C, dated 1 November 1961.

PAGES 1 and 2

* 2.1, under the heading FEDERAL for specifications delete the following documents:

"PPP-B-576 Box, Wood, Cleated, Veneer Paper Overlaid
PPP-B-591 Boxes, Fiberboard, Wood-Cleated
PPP-T-76 Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant".

* under the same heading add the following documents:

"NN-P-71 Pallets, Material Handling, Wood Stringer Construction, 2 Way and 4 Way (Partial)
QQ-S-781 Strapping, Steel, and Seals
PPP-B-640 Box, Fiberboard, Corrugated, Triple-wall
PPP-F-320 Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760 Strapping, Nonmetallic, (and Connectors)
PPP-T-97 Tape, Pressure-sensitive Adhesive, Filament Reinforced".

PAGE 2

* 2.1, under the heading MILITARY for specifications delete the following document:

"MIL-L-10547 Liners, Case, Waterproof".

* under the heading MILITARY for standards add the following document:

"MIL-STD-147 Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners or Pallet Type Base".

FSC 6135

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:37Z
Check the source to verify that this is the current version before use.

PAGE 6

3.7.2.1, under the heading, Capacity and voltage tests (minimum time to reach test end voltage), make the following changes:

Delete 90 minutes for "initial and delayed normal temperature discharge", and substitute 110 minutes.

Add new requirement, "Flight similitude discharge . . . 100 minutes".

PAGE 11

4.3.1, following this paragraph add new subparagraph as follows:

"4.3.1.1 Test Chamber Conditions. During low temperature discharge tests and the low temperature phase of the high-low temperature discharge tests, the air motion within a proximity of 2 inches to 6 inches of the battery shall be maintained in the range of 40 to 100 feet per minute."

4.5, delete and substitute:

"4.5 Preproduction inspection. Preproduction inspection shall consist of the tests specified in tables II, IV, and V, for test groups I and II, and in table III, for test groups I, II, and IV."

4.6.1, delete and substitute:

"4.6.1 Lot. All batteries of the same type designation offered for delivery during any one month on any one contract or order, shall be considered a lot (quantity on release) for purposes of sampling and inspection."

"4.6.1.1 Sublot. At the supplier's option, sampling and inspection may be performed more frequently on a smaller-than-lot (sublot) basis. The group A sample size for a sublot shall be determined in the same manner as that for a lot, using the size of the sublot submitted and the inspection level specified (see 4.6.2.1). The group B sample size for a sublot shall be determined in the same manner as that for a lot, using the size of the sublot submitted and the inspection level specified (see 4.6.3.1). The group C sample size for a sublot shall have the same ratio to the lot sample size that is specified in table I (see 4.6.4) as the sublot has to the lot (quantity of release)."

4.6.2.1, second sentence: Delete and substitute: "The acceptable quality level shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects."

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:37Z
Check the source to verify that this is the current version before use.

4.6.3.1, line 3: Delete "section B" and substitute "section C", to end of paragraph add: "the acceptable quality level shall be 6.5 percent defective".

* 4.6.4.1, add the following to end of paragraph: "Sample batteries shall be shipped to the following address:

US Army Electronics Command
Product Assurance Directorate
ATTN: DRSEL-PA-OD
Building T-292, Squier Area
Fort Monmouth, New Jersey 07703"

4.6.4.2, line 3: Delete "test groups I and III in the order shown".

4.6.4.3, delete and substitute:

"4.6.4.3 All sample batteries shall be subjected to the visual and dimensional examination in inspection group I. The sample batteries shall then be divided between inspection groups II and III. Two batteries shall be assigned to each of the capacity voltage tests of inspection group II, and the remainder placed in group III. After the storage period, inspection group III shall be further divided so that approximately equal quantities of batteries shall be activated and subjected to the tests specified therein."

4.6.4.4, following this paragraph add new paragraphs as follows:

"4.6.5 Group D inspection. (Only applicable to BA-259/AM) Group D inspection shall consist of the flight similitude discharge test and is to be performed by the government under the conditions as specified herein."

"4.6.5.1 Sampling. In addition to the samples in accordance with paragraph 4.6.4, 5 additional batteries shall be selected at random from the quantity on release by the government representative."

4.7.3, line 2: Delete "(see 4.8.4)" and substitute "(3.4 and 3.7)".

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:37Z
Check the source to verify that this is the current version before use.

4.7.5.2, add new test group to table III, and footnote, as follows:

Inspection group	Inspection	Storage time at 70° ± 2°F. (21.1°±1.1°C.) and 50± 5% RH	Temperature for discharge through resistance load, after activation
IV	Flight simulitude discharge	As applicable	See [2]

[2] After the battery has been removed from storage, activated in accordance with manufacturer's instructions, and has reached the minimum initial voltage, the battery shall then be inserted in radiosonde AN/AMT-4, AN/AMT-12, or a substitute box having the same insulating properties as the radiosonde. The radiosonde or the substitute box, with the activated battery mounted therein, shall be placed in a test chamber at ambient room temperature and ambient atmosphere pressure. Simultaneously with the beginning of the discharge, the pressure and the temperature shall be reduced in accordance with table IIIa. Temperature, pressure and voltage of each unit shall be read at 5 minute intervals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:37Z
Check the source to verify that this is the current version before use.

Table IIIa. Time, Temperature, Pressure for Simulated Flight Testing

TIME From instant of simulated-flight initiation (Tolerance: ± 1 minute) Minutes	ALTITUDE (simulated) Above sea level Feet	PRESSURE In the test chamber (Tolerance: ±10% of the listed pressure) Inches of Hg.	TEMPERATURE In the test chamber (Tolerance: ±5°F.) °F
0.0	0	29.92 (Approx)	+59.0
5.0	5,000	24.90	+41.2
10.0	10,000	20.58	+23.3
15.0	15,000	16.89	+ 5.5
20.0	20,000	13.75	-12.3
25.0	25,000	11.10	-30.2
30.0	30,000	8.89	-48.0
35.0	35,000	7.04	-65.8
40.0	40,000	5.54	-69.7
45.0	45,000	4.36	-69.7
50.0	50,000	3.42	-69.7
55.0	55,000	2.69	-69.7
60.0	60,000	2.12	-69.7
65.0	65,000	1.67	-69.7
70.0	70,000	1.31	-69.7
75.0	75,000	1.03	-69.7
80.0	80,000	0.81	-69.7
85.0	85,000	0.64	-64.8
90.0	90,000	0.50	-56.6
95.0	95,000	0.40	-48.3
100.0	100,000	0.32	-40.1

* 4.7.5.4, following this paragraph add a new paragraph as follows:

"4.8 Quality Conformance Inspection of Packaging. Packing shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5. The rough handling test will be performed only when invoked in the bid request or contract [See 6.1(e)]."

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:37Z
Check the source to verify that this is the current version before use.

PAGES 14 thru 17

* 5.1 thru 5.3, delete the entire Section 5 and replace with the following paragraphs:

5. PACKAGING

5.1 Preservation. Preservation shall be level A or B as specified (see 6.1(d)).

5.1.1 Level A.

5.1.1.1 Cleaning. Each battery shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each battery shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Each battery shall be individually unit packed in accordance with MIL-P-116, submethod 1A-5 as follows: Cushion by wrapping with cushioning conforming to PPP-C-843, type II, class B, 1/2 inch thick. Place the cushioned item into a can conforming to type III of PPP-C-96. Plan B exterior nonmetallic coating shall be required.

5.1.2 Level B.

5.1.2.1 Cleaning, drying, preservative application and unit packing. Each battery shall be cleaned, dried, preserved and unit packed as specified in 5.1.1.

5.2 Packing. Packing shall be level A or B as specified (see 6.1(d)). Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of batteries, as specified in table VI, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4; or PPP-B-585, class 3, style 2 or 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:37Z
Check the source to verify that this is the current version before use.

accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

TABLE VI PACKING QUANTITY

TYPE	QUANTITY
BA-253/U	72
BA-259/AM	24
BA-292/AM	50
BA-316/AM	12

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet conforming to MIL-STD-147, load type I. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.2 Level B. A quantity batteries, as specified in table VI, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the containers shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:37Z
Check the source to verify that this is the current version before use.

5.2.2.1 Unitization. Palletization shall be required when: containers specified in 5.2.2 do not require pallet mounting; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.2, shall be placed on a pallet, conforming to MIL-STD-147, load type I. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.3 Marking

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129. In addition, the top of each unit package shall be marked, "CAUTION--DO NOT OPEN UNTIL BATTERY IS TO BE ACTIVATED". The size of the lettering shall be a minimum of 1/4 inch.

PAGE 17

6.1, in item (d) delete "Level of preservation, packaging and packing required (see 5.1 and 5.2)" and substitute "Level A or Level B preservation and packing (see section 5)." Also add an item "(e) When rough handling test is required."

6.2.2: delete the "value of adjustment formula and substitute:

"Value of adjustment = $\frac{M(MPI-API)"}{(MPI)}$

PAGES 17 and 18

6.2.3: delete the "value of adjustment" formula and substitute:

"Value of adjustment = $\frac{N(MPI-WA)"}{(WA)}$;

last sentence: Delete and substitute: "No adjustment should result if the computed WA is equal to the MPI."

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:37Z
Check the source to verify that this is the current version before use.

* End of section 6, add the following paragraph:

"6.3 Environmental. Environmental pollution protection measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods."

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army - EL

Preparing activity:
Army - EL

(Project 6135-A141)

☆ U. S. GOVERNMENT PRINTING OFFICE: 1977-703-020/1469

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:37Z
Check the source to verify that this is the current version before use.